

LED1 shows power-on status of the circuit, whereas LED2 through LED5 are used as indicators to show status of the four relays (RL-RL4). Power to the relay board is provided separately from a +9V battery.

Software

1. Arduino IDE is used for programming Arduino Uno. The latest IDE

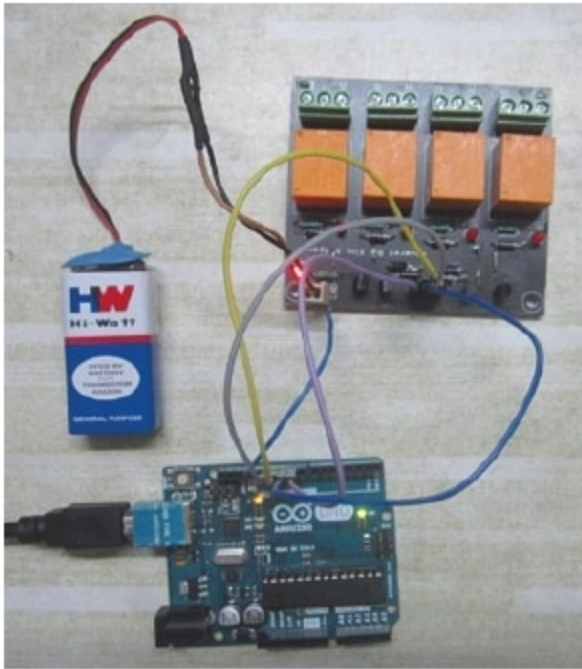


Fig. 3: Prototype of relay board



Fig. 4: Block diagram for controlling equipment

can be downloaded for free from Arduino's official website. After downloading Arduino IDE, install it. Note the directory of Arduino IDE installation.

2. Download Legacy MATLAB and Simulink Support for Arduino package from MathWorks website. Extract the compressed folder named ArduinoIO. From ArduinoIO folder, copy pde folder and paste it in C:\Program Files (or Program files x86)\Arduino\libraries. (Path may be different depending on the installation directory of the IDE.)

After pasting pde folder in the correct location, open Arduino IDE. If you have pasted the folder in the correct location, you will find pde in File→Examples→pde. Open the code from File→Examples→pde→adioes.

3. Connect Arduino Uno board to the PC. From Device Manager, note the COM port at which Arduino Uno is installed. From Tools in Arduino IDE, select the board as Arduino Uno and select the COM port number noted earlier. Upload adioes code to Arduino Uno by clicking Upload in the IDE.

4. Copy the entire contents of the extracted ArduinoIO folder to a folder in My Documents (for Windows PC).

5. Our GUI application program

PARTS LIST	
<i>Semiconductors:</i>	
Board1	- Arduino Uno board
D1-D8	- 1N4007, rectifier diode
LED1-LED5	- 5mm LED
T1-T4	- BC547 npn transistor
<i>Resistors (all 1/4-watt, ±5% carbon):</i>	
R1, R6-R9	- 1-kilo-ohm
R2-R5	- 10-kilo-ohm
<i>Miscellaneous:</i>	
CON1	- 9V DC power supply
CON2-CON5	- 3-pin connectors for load
RL1-RL4	- 9V, single-changeover relay
	- USB cable for Arduino

has been developed in R2014a version of MATLAB. After installing this version of MATLAB on the PC, open install_arduino.m file present in the directory where you have copied the contents of ArduinoIO folder. Now, run install_arduino.m file. This code should install correctly and save the path of Arduino support package.

6. Open the source code file for this project, titled relay_control.m. Edit a = arduino ('COM9') line with the COM port number in the PC where Arduino Uno has been installed. After running the project file, MATLAB will try to communicate with the board. After successful communication is established, you will be able to control the equipment by clicking on the appropriate pushbutton in the GUI.

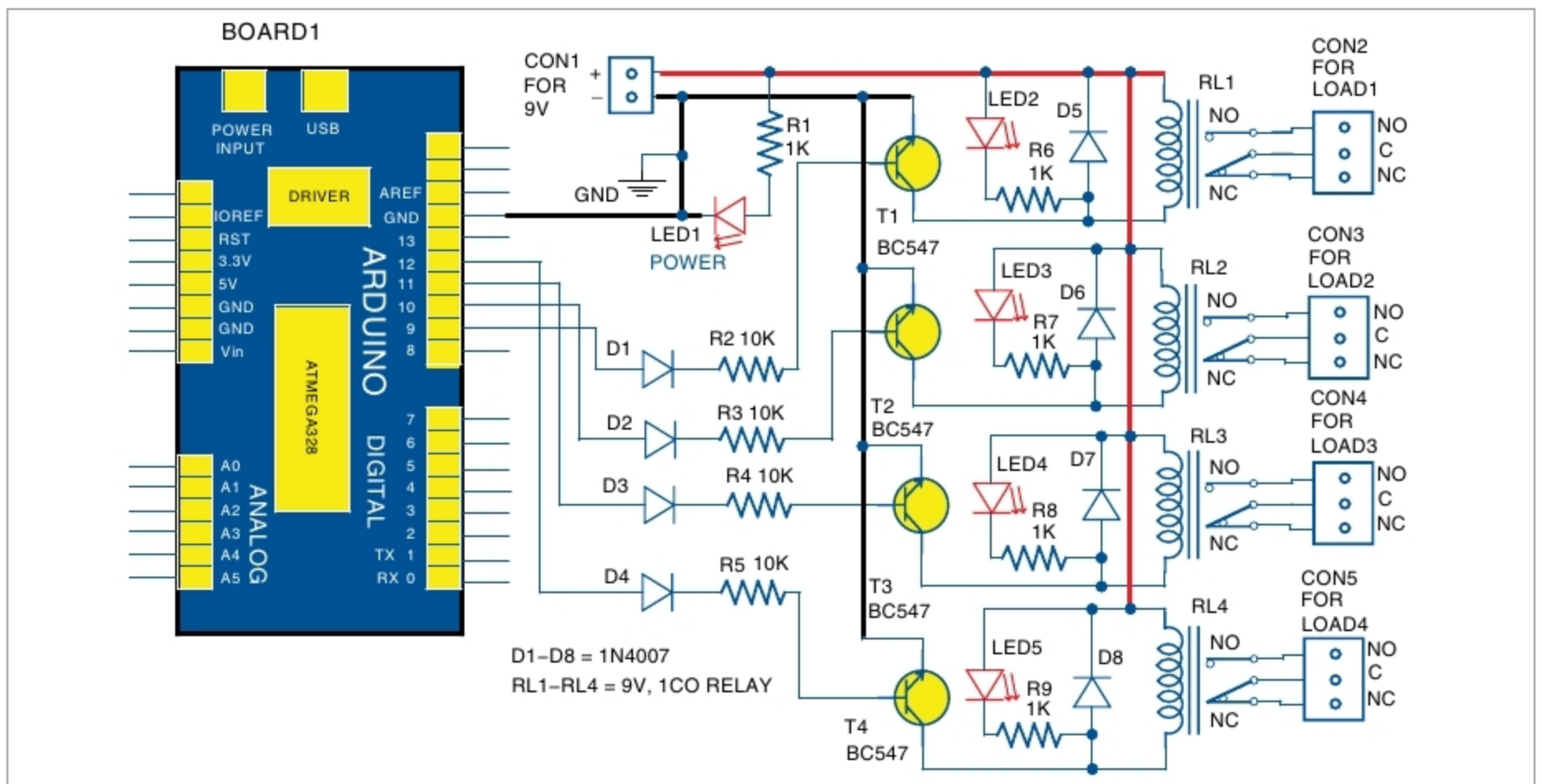


Fig. 5: Circuit diagram for the equipment controller